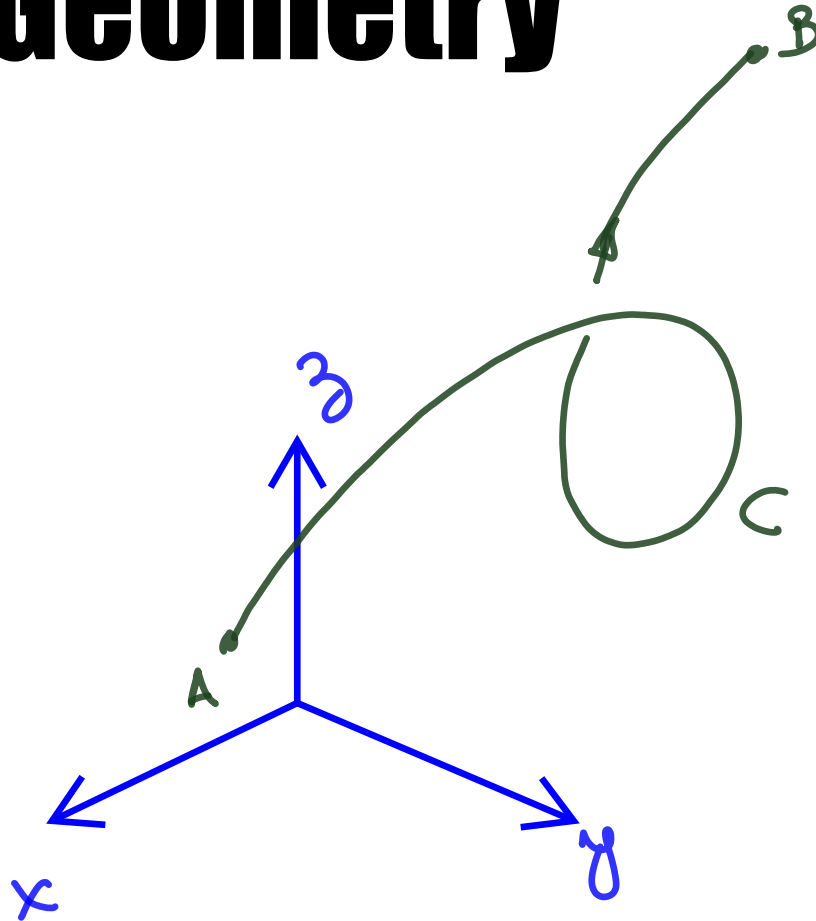


3D Analytic Geometry

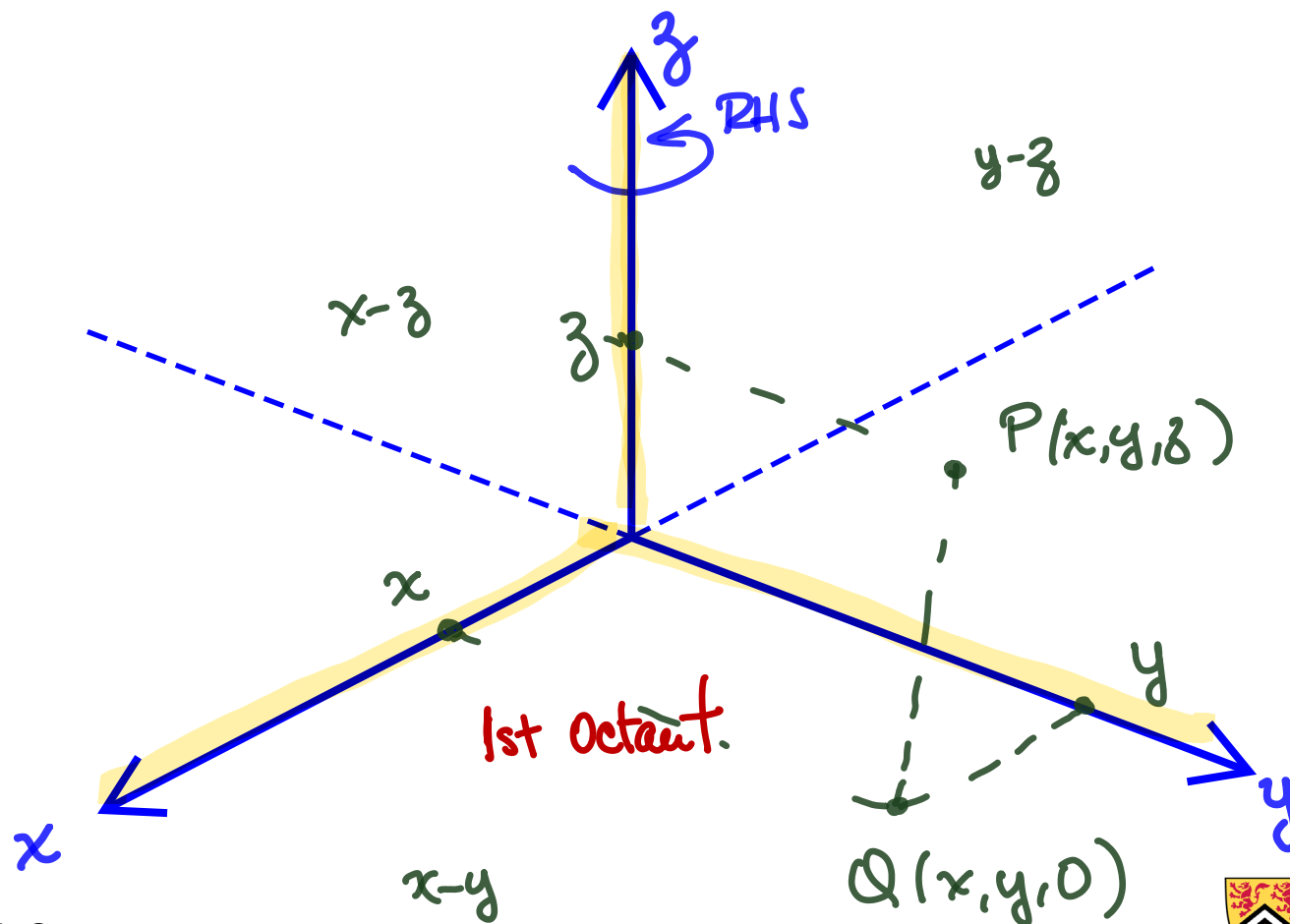
Trim Ch 11, S. 11.1



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Rectangular Coordinates in Space

We use three mutually perpendicular coordinate axes x , y and z , and three coordinate planes $x - y$, $x - z$, and $y - z$ to locate a point in space.



Writing Functions in 3D Rectangular Coordinates

What happens to the math we know in $\mathbb{R}^2$ when we step into $\mathbb{R}^3$?

1. Distance between two points P_1 and P_2 :

$$d(P_1 P_2) = \sqrt{\underbrace{(x_2 - x_1)^2 + (y_2 - y_1)^2}_{\mathbb{R}^2} + (z_2 - z_1)^2}$$

2. Circle centered at (a, b) and radius r :

$$\underbrace{(x-a)^2 + (y-b)^2}_{\mathbb{R}^2} + \underbrace{(z-c)^2}_{\mathbb{R}^3} = r^2$$

This looks simple enough ... but we need to be cautious.

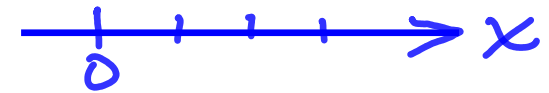
Not everything from $\mathbb{R}^2$ carries over neatly into $\mathbb{R}^3$!

Example:

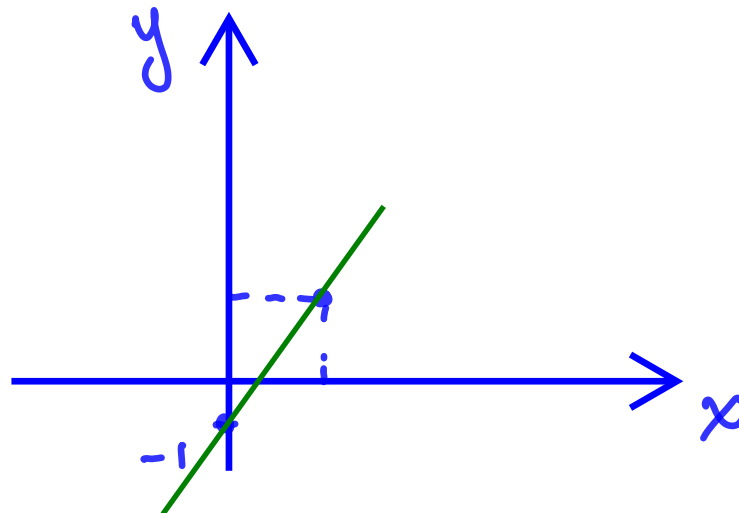
Graph the equation $y = 2x - 1$ in $\mathbb{R}$, $\mathbb{R}^2$ and $\mathbb{R}^3$

Solution:

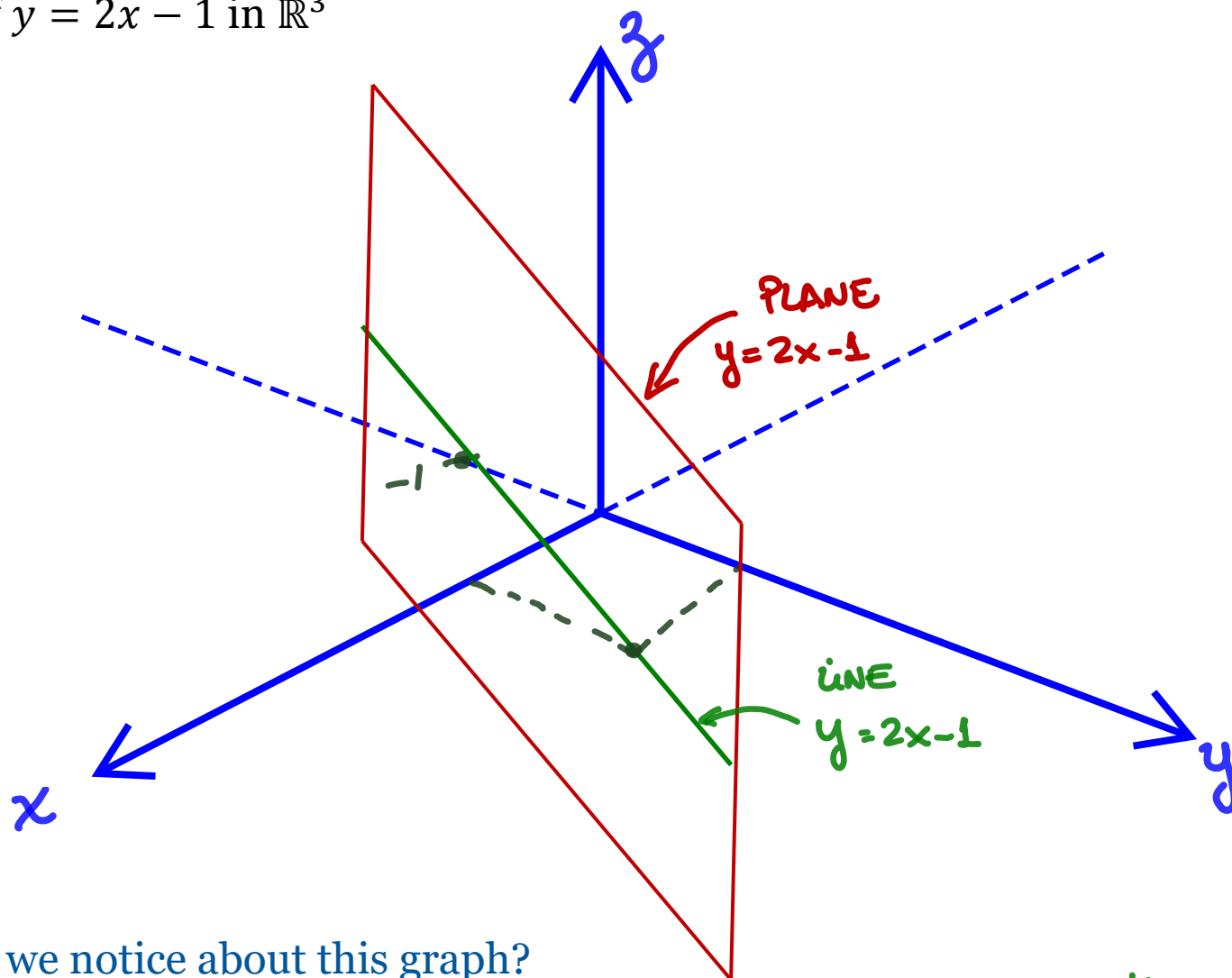
Graph in $\mathbb{R} \rightarrow$ We can't graph it in $\mathbb{R}$



Graph in $\mathbb{R}^2$



Graph of $y = 2x - 1$ in $\mathbb{R}^3$

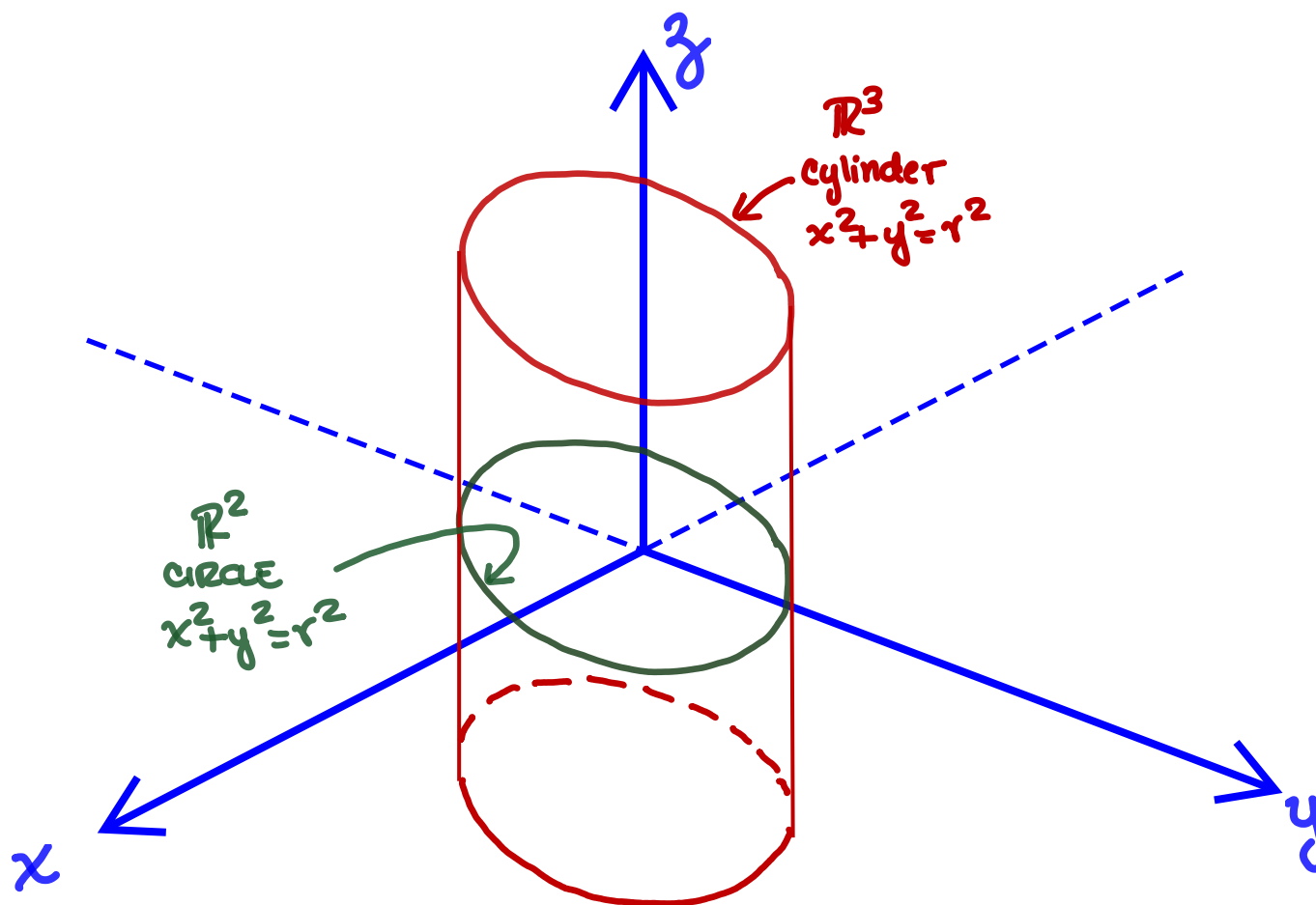


What do we notice about this graph?

Intersection of Plane $y = 2x - 1$ with x - y plane is a line.

Question:

What does the graph of the equation $x^2 + y^2 = r^2$ represents in $\mathbb{R}^3$?



Summary

- We defined the three-dimensional space ($\mathbb{R}^3$) using the rectangular coordinate system.
- We have illustrated that functions describing curves in $\mathbb{R}^2$, describe surfaces in $\mathbb{R}^3$
- The intersection of two simultaneous surfaces in $\mathbb{R}^3$, specify a curve in the three-dimensional space

How can we represent a curve in $\mathbb{R}^3$?



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Thank You